Absolutely. You need **normal, human-readable mathematical formulas** that you can paste directly into Word/Google Docs, export as PDF, and then feed into your converter.

Here is a good **test set from simple to complex**.

Basic Algebra

1. $x + y = 10$
2. $2x + 5 = 15$
3. $x^2 + 5x + 6 = 0$
4. $a^2 + b^2 = c^2$
5. $(x + y)^2 = x^2 + 2xy + y^2$
6. $x = (-b \pm \sqrt{}(b^2 - 4ac)) / 2a$
7. $(a + b) / (c + d) = (a + b) / (c + d)$
8. $(x^2 - 9) / (x - 3) = x + 3$

Fractions and Roots

9. $1/2 + 1/3 = 5/6$
10. $\sqrt{}(x^2 + y^2) = z$
11. $\sqrt{}(a + \sqrt{}b) = c$
12. $(x + 1)/(x - 1) + (x - 1)/(x + 1) = 2(x^2 + 1)/(x^2 - 1)$
13. $1/(1 + 1/(1 + x)) = (1 + x)/(2 + x)$

Powers and Logarithms

14. $a^m \times a^n = a^{m+n}$
15. $(a^m)^n = a^{mn}$
16. $\log(ab) = \log(a) + \log(b)$
17. $\log(a/b) = \log(a) - \log(b)$
18. $\log_a(x) = \ln(x) / \ln(a)$
19. $e^x = 1 + x + x^2/2! + x^3/3! + \cdots$

Sequences and Series

20. $1 + 2 + 3 + \cdots + n = n(n + 1)/2$
21. $1^2 + 2^2 + 3^2 + \cdots + n^2 = n(n + 1)(2n + 1)/6$
22. $1 + r + r^2 + \cdots + r^n = (r^{n+1} - 1)/(r - 1)$
23. $\Sigma_{i=1}{}^n\ i = n(n + 1)/2$
24. $\Sigma_{i=1}{}^n\ i^2 = n(n + 1)(2n + 1)/6$
25. $\Pi_{i=1}{}^n\ i = n!$

Trigonometry

26. $\sin^2\theta + \cos^2\theta = 1$
27. $\tan \theta = \sin \theta / \cos \theta$
28. $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$

29. $\cos(\alpha + \beta) = \cos \alpha \cos \beta - \sin \alpha \sin \beta$
30. $\sin 2\theta = 2 \sin \theta \cos \theta$
31. $\cos 2\theta = \cos^2\theta - \sin^2\theta$
32. $1 + \tan^2\theta = \sec^2\theta$

Calculus

33. $dy/dx = 2x$
34. $d/dx\ (x^n) = nx^{n-1}$
35. $d/dx\ (\sin x) = \cos x$
36. $d/dx\ (\cos x) = -\sin x$
37. $\int x^2\, dx = x^3/3 + C$
38. $\int \sin x\, dx = -\cos x + C$
39. $\int_0^1 x^2\, dx = 1/3$
40. $\lim_{x \to 0} (\sin x)/x = 1$
41. $d^2y/dx^2 + 4y = 0$

More Complex Calculus

42. $\int_0^\infty e^{-x}\, dx = 1$
43. $\int_0^1 x^n\, dx = 1/(n + 1)$
44. $d/dx\ [x^2 \sin x] = 2x \sin x + x^2 \cos x$
45. $\int x e^x\, dx = e^x(x - 1) + C$
46. $\partial^2u/\partial x^2 + \partial^2u/\partial y^2 = 0$
47. $\partial u/\partial t = \alpha\, \partial^2u/\partial x^2$

Matrices

48.

$$\begin{vmatrix} 1 & 2 \\ 3 & 4 \end{vmatrix}$$

49.

$$A = \begin{vmatrix} a & b \\ c & d \end{vmatrix}$$

50.

$$A = \begin{vmatrix} 1 & 2 & 3 \\ 4 & 5 & 6 \\ 7 & 8 & 9 \end{vmatrix}$$

51.

$\det(A) = ad - bc$

52.

$$A^{-1} = 1/(ad - bc) \begin{vmatrix} d & -b \\ -c & a \end{vmatrix}$$

Probability and Statistics

53. $P(A \cup B) = P(A) + P(B) - P(A \cap B)$
54. $P(A \mid B) = P(A \cap B) / P(B)$
55. $E(X) = \Sigma\, x_i P(X = x_i)$
56. $Var(X) = E(X^2) - [E(X)]^2$
57. $\sigma^2 = \Sigma(x_i - \mu)^2 / N$
58. $z = (x - \mu) / \sigma$
59. $f(x) = 1/(\sigma\sqrt{(2\pi)})\ e^{-(x-\mu)^2/2\sigma^2}$

Coordinate Geometry

60. Distance = $\sqrt{[(x_2 - x_1)^2 + (y_2 - y_1)^2]}$
61. $m = (y_2 - y_1)/(x_2 - x_1)$
62. $y - y_1 = m(x - x_1)$
63. $x^2 + y^2 = r^2$
64. $(x - h)^2 + (y - k)^2 = r^2$
65. $Ax + By + C = 0$

Vectors

66. $\mathbf{a} \cdot \mathbf{b} = |a||b| \cos\theta$
67. $|a \times b| = |a||b| \sin\theta$
68. $\mathbf{a} = a_1 i + a_2 j + a_3 k$
69. $|a| = \sqrt{(a_1^2 + a_2^2 + a_3^2)}$
70. $a \cdot b = a_1 b_1 + a_2 b_2 + a_3 b_3$

Differential Equations

71. $dy/dx + y = e^x$
72. $d^2y/dx^2 + 3dy/dx + 2y = 0$
73. $y'' + \omega^2 y = 0$
74. $dy/dx = ky$
75. $y = Ce^{kx}$

Physics/Formulas

76. $F = ma$
77. $E = mc^2$
78. $v = u + at$
79. $s = ut + \frac{1}{2}at^2$
80. $v^2 = u^2 + 2as$
81. $p = mv$
82. $E = \frac{1}{2}mv^2$
83. $V = IR$

84. $P = VI$
85. $Q = mc\Delta T$

Advanced but still normal-looking formulas

86. **Quadratic formula**

$x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$

87. **Binomial theorem**

$(a + b)^n = \Sigma_{r=0}^n C(n,r)a^{n-r}b^r$

88. **Taylor series**

$f(x) = f(a) + f'(a)(x - a) + f''(a)(x - a)^2/2! + \cdots$

89. **Euler's formula**

$e^{ix} = \cos x + i \sin x$

90. **Normal distribution**

$f(x) = 1/(\sigma\sqrt{(2\pi)}) \exp(-(x - \mu)^2/(2\sigma^2))$

91. **Bayes' theorem**

$P(A | B) = P(B | A)P(A) / P(B)$

92. **Fourier transform**

$F(\omega) = \int_{-\infty}^{\infty} f(t)e^{-i\omega t}\, dt$

93. **Heat equation**

$\partial u/\partial t = \alpha^2\, \partial^2 u/\partial x^2$

94. **Wave equation**

$\partial^2 u/\partial t^2 = c^2\, \partial^2 u/\partial x^2$

95. **Schrödinger equation**

$i\hbar\, \partial\psi/\partial t = -(\hbar^2/2m)\nabla^2\psi + V\psi$

For your PDF test, I recommend these 15

If your goal is specifically to test whether **PDF → image → pix2tex → LaTeX → MathML → JATS XML** works correctly, put these together in one PDF:

1. $x^2 + 5x + 6 = 0$
2. $x = (-b \pm \sqrt{(b^2 - 4ac)}) / 2a$
3. $(a + b)^2 = a^2 + 2ab + b^2$
4. $1 + 2 + 3 + \cdots + n = n(n + 1)/2$
5. $\Sigma_{i=1}^{n} i^2 = n(n + 1)(2n + 1)/6$
6. $\sin^2\theta + \cos^2\theta = 1$
7. $\int_0 \infty\ e^{-x}\ dx = 1$
8. $d^2y/dx^2 + 4y = 0$
9. $\partial^2u/\partial x^2 + \partial^2u/\partial y^2 = 0$
10. $A^{-1} = 1/(ad - bc)\ [d\ -b;\ -c\ a]$
11. $P(A \mid B) = P(A \cap B)/P(B)$
12. $E(X) = \Sigma x_i P(X = x_i)$
13. $e^{ix} = \cos x + i \sin x$
14. $f(x) = 1/(\sigma\sqrt{(2\pi)})\ \exp(-(x - \mu)^2/(2\sigma^2))$
15. $i\hbar\ \partial\psi/\partial t = -(\hbar^2/2m)\nabla^2\psi + V\psi$

These will give you a good range of **fractions, roots, Greek characters, superscripts, subscripts, summations, integrals, matrices, derivatives, limits, and nested expressions** to evaluate the conversion quality.